

Software Requirements Specification (SRS)

Product: Vaccination Cohort & Dose Scheduling System (Problem Statement #17)

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Structure: IEEE Std 830-1998

1. Introduction

1.1 Purpose

This document states what the Vaccination Cohort & Dose Scheduling System must do and how well it must do it. It is written for the people who design, build, test and grade the system.

1.2 Scope

The product is a public health administration platform. It will:

- register citizens and place each one in a vaccination cohort (phase)
- stop a citizen from booking the next dose before the minimum interval has passed
- let citizens find a centre by pincode and date and book a slot within the centre's capacity
- let Vaccination Officers check citizens in and record each dose
- issue a digitally signed QR vaccination certificate after the final dose, and verify it on scan

Out of scope: vaccine stock and cold-chain management, payments, and adverse-event reporting.

1.3 Definitions and abbreviations

Term	Meaning
Cohort / phase	A group of citizens who become eligible together (for example healthcare workers first)
Dose interval	Minimum number of days between two doses (28 days by default)
Slot	A bookable place at one centre on one date, limited by daily capacity
QR certificate	Vaccination proof whose QR code carries citizen ID, dose dates and a digital signature
FR / NFR	Functional / non-functional requirement
RTM	Requirements traceability matrix
PII	Personally identifiable information
RBAC	Role-based access control

1.4 References

1. Problem Statement #17, SE Lab 1 handout, PES University
2. IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
3. 1-RE/FR_and_NFR.md and 1-RE/RTM_Table.md
4. 2-Architectural_Diagram/Architecture_Specification.md

1.5 Overview

Section 2 describes the product and its users. Section 3 lists the specific requirements.

2. Overall description

2.1 Product perspective

A new, self-contained system built as microservices behind one API gateway. Citizens use a web or mobile portal. Officers use a console at the vaccination centre. Each service keeps its own database.

2.2 Product functions

Registration and cohort assignment, dose-interval enforcement, slot search and booking, dose recording, certificate generation, certificate verification, and booking notifications.

2.3 User classes

User	What they do	Skill assumed
Citizen Registrant	Registers, books slots, downloads the certificate	Basic smartphone use, may be a first-time user
Vaccination Officer	Checks citizens in, records doses, scans certificates	Trained on the console, works under time pressure

2.4 Operating environment

- Citizen Portal: current mobile and desktop browsers
- Officer Console: browser on a centre laptop or tablet with a camera for QR scanning
- Backend: containerised services on a cloud platform, one relational database per service

2.5 Design and implementation constraints

- Dose rules (interval days, required doses, cohort criteria) must be configuration, not hard-coded
- All traffic uses TLS 1.2 or later, and stored personal data uses AES-256 (NFR-002)
- The certificate signing key is held only by the Certificate Service
- Certificate verification must work offline, so the QR payload carries everything needed to check it

2.6 Assumptions and dependencies

- Each citizen has one valid government ID and a reachable phone number or email
- Centres publish their daily capacity before booking opens
- An SMS or email provider is available for notifications

3. Specific requirements

3.1 External interface requirements

Interface	Requirement
User	Citizen Portal: booking in a few clear steps, readable on a phone. Officer Console: one-screen check-in form with required fields marked
Hardware	Camera on the officer device for QR scanning
Software	SMS or email gateway for notifications, relational databases, PDF generation for certificates
Communication	REST over HTTPS between clients and the gateway, REST or asynchronous messages between services

3.2 Functional requirements

3.2.1 Registration and cohort assignment (FR-002)

- **Input:** name, age, government ID, comorbidity flag, occupation category.
- **Processing:** validate the fields, reject duplicate IDs and under-age citizens, assign the earliest cohort the citizen qualifies for (occupation, then age or comorbidity).

- **Output:** citizen profile with its cohort.

3.2.2 Dose interval enforcement (FR-001)

- **Input:** citizen ID and the date of the slot being requested.
- **Processing:** compare the slot date with the date of the previous dose. Block when the gap is under the minimum interval (28 days).
- **Output:** booking allowed, or a message saying how many days remain.

3.2.3 Centre search and slot booking (FR-003)

- **Input:** pincode, date, chosen slot.
- **Processing:** list slots with seats left. On booking, check capacity and take the seat in a single atomic step so the last seat cannot be given twice.
- **Output:** booking confirmation, and the remaining count goes down by one.

3.2.4 Check-in and dose recording (FR-004)

- **Input:** booking, vaccine brand, batch/lot number, date, officer ID.
- **Processing:** reject a blank batch number. Save the dose record. Move the citizen status from "Slot Booked" to "Dose Administered".
- **Output:** updated vaccination record.

3.2.5 Certificate generation (FR-005)

- **Input:** citizen with the required number of doses recorded.
- **Processing:** build the certificate data (citizen ID, dose dates), sign it, encode it as a QR code. Never issue before the final dose.
- **Output:** signed QR certificate, downloadable as a PDF.

3.2.6 Certificate verification (FR-006)

- **Input:** scanned QR payload.
- **Processing:** check the signature against the payload.
- **Output:** "Valid" with the citizen's name and dose dates, or "Invalid".

3.2.7 Notifications (FR-007)

- **Input:** a successful booking.
- **Processing:** queue one confirmation at once and one reminder 24 hours before the slot.
- **Output:** SMS or email to the citizen.

3.3 Non-functional requirements

ID	Quality	Requirement	Measure
NFR-001	Performance & Security	Verify a signed QR certificate offline or online	Under 150 ms under simulated peak load
NFR-002	Security & Privacy	Encrypt personal and health data at rest and in transit, officers only	TLS 1.2+, AES-256, RBAC confirmed by security review
NFR-003	Scalability	Handle a rollout rush of bookings	5,000 concurrent requests, 95% answered within 3 s, zero double-bookings
NFR-004	Availability	Recording and verification keep working if booking is down	Both actions succeed with the booking service stopped
NFR-005	Auditability	Know who recorded each dose and when	Officer ID and date on every record, changes logged

3.4 Logical database requirements

Store	Owned by	Holds
Citizen DB	Citizen Registry Service	Profiles, cohort, government ID (encrypted)
Booking DB	Slot Booking Service	Centres, slots, capacity, bookings
Records DB	Vaccination Record Service	Dose records, interval rules, audit entries
Certificate Store	Certificate Service	Issued certificates and signing key reference

No service reads another service's database. It asks through that service's interface.

Work Breakdown Structure (WBS)

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The project is split into six phases. Phases 1 to 3 match Labs 1 to 3. Phase 4 uses the same seven stories and story points as the Jira Scrum project (Scrum_BPS#17 , 39 points in total).

1. WBS tree

Vaccination Cohort & Dose Scheduling System

- 1. Requirements
 - 1.1 Analyse the problem statement and actors
 - 1.2 Write the FR and NFR table
 - 1.3 Draw the use-case diagram and one use-case flow
- 2. Agile planning
 - 2.1 Kanban board: requirements, epics, user stories
 - 2.2 Scrum backlog: priorities and story points
 - 2.3 Sprint simulation and burndown chart
 - 2.4 Bug tracker: log and triage defects
- 3. Architecture and design
 - 3.1 Compare Layered, Client-Server and Microservices
 - 3.2 Draw the UML component diagram
 - 3.3 Specify the interfaces between components
 - 3.4 Write the one-page justification
- 4. Implementation
 - 4.1 Citizen registration and cohort assignment
 - 4.2 Dose interval lock
 - 4.3 Centre search and slot booking
 - 4.4 Dose administration check-in
 - 4.5 Digital certificate generation
 - 4.6 Certificate verification
 - 4.7 Citizen data security
 - 4.8 Booking notifications
- 5. Testing
 - 5.1 Unit tests for the core rules
 - 5.2 Fix and retest the logged defects
 - 5.3 Load, security and failover tests
- 6. Release
 - 6.1 Package and deploy the services
 - 6.2 Pilot at one vaccination centre
 - 6.3 Documentation and handover

2. Work package dictionary

WBS	Work package	Requirement	Jira story	Story points	Output	Status
1.1	Analyse problem statement	all			Actors and scope	Done (Lab 1)
1.2	FR and NFR table	all			Requirements table	Done (Lab 1)
1.3	Use-case model	all			Use-case diagram, flow spec	Done (Lab 1)
2.1	Kanban board	baseline 7			3 epics, 7 stories with tasks	Done (Lab 2)

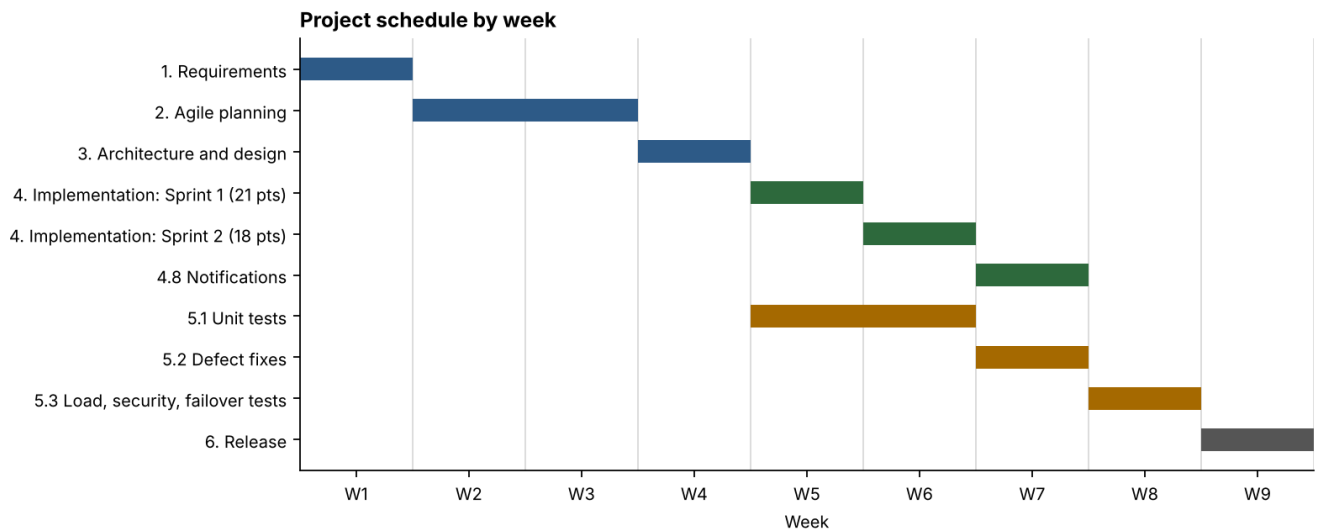
WBS	Work package	Requirement	Jira story	Story points	Output	Status
2.2	Scrum backlog	baseline 7			Prioritised backlog, 39 points	Done (Lab 2)
2.3	Sprint simulation	baseline 7			2 sprints, burndown chart	Done (Lab 2)
2.4	Bug tracker	baseline 7			7 defects logged (BBPS-1 to 7)	Done (Lab 2)
3.1	Style comparison	NFR-001 to 004			Comparison table	Done (Lab 3)
3.2	Component diagram	all			8 components, 10 interfaces	Done (Lab 3)
3.3	Interface specification	all			Architecture specification	Done (Lab 3)
3.4	Justification	NFR-001 to 004			One-page document	Done (Lab 3)
4.1	Registration and cohort	FR-002	SBPS-2	5	Citizen Registry rules	Core rules built
4.2	Dose interval lock	FR-001	SBPS-1	3	Eligibility check	Core rules built
4.3	Slot booking	FR-003	SBPS-3	8	Booking engine	Core rules built
4.4	Dose check-in	FR-004, NFR-005	SBPS-4	5	Dose recording	Core rules built
4.5	Certificate generation	FR-005	SBPS-5	8	Signed QR payload	Core rules built, PDF export pending
4.6	Certificate verification	FR-006, NFR-001	SBPS-6	5	Verification check	Core rules built
4.7	Data security	NFR-002	SBPS-7	5	Encryption, RBAC	Planned
4.8	Notifications	FR-007	not in backlog	not estimated	Confirmation and reminder	Confirmation built, reminder pending
5.1	Unit tests	FR-001 to FR-007			15 automated tests	Done, 15 of 15 pass
5.2	Defect fixes	baseline 7	BBPS-1 to 7		Regression tests	6 of 7 have a regression test
5.3	Load, security, failover tests	NFR-001 to 004			Test reports	Planned
6.1	Deploy	all			Running services	Planned
6.2	Pilot	all			Pilot feedback	Planned
6.3	Handover	all			User and operations guides	Planned

"Core rules built" means the business logic exists and is unit tested in `vaccination_core_engine.py`. The user interfaces, databases and deployed services are not built.

Story points by sprint (from Jira)

Sprint	Stories	Work packages	Points
Sprint 1	SBPS-1, SBPS-2, SBPS-3, SBPS-4	4.1, 4.2, 4.3, 4.4	21
Sprint 2	SBPS-5, SBPS-6, SBPS-7	4.5, 4.6, 4.7	18
Total	7 stories		39

3. Schedule



Week	Work
1	Phase 1: requirements
2 to 3	Phase 2: agile planning
4	Phase 3: architecture and design
5	Sprint 1: work packages 4.1 to 4.4, unit tests alongside
6	Sprint 2: work packages 4.5 to 4.7, unit tests alongside
7	4.8 notifications, 5.2 defect fixes
8	5.3 load, security and failover tests
9	Phase 6: deploy, pilot, handover

Milestones: M1 requirements agreed (end of week 1), M2 backlog ready (week 3), M3 architecture approved (week 4), M4 all stories done (week 6), M5 tests passed (week 8), M6 pilot live (week 9).

Dependencies: 4.3 needs 4.1 and 4.2 (booking checks the cohort and the interval). 4.5 needs 4.4 (a certificate needs dose records). 4.6 needs 4.5. 5.3 needs every package in phase 4.

4. Risks

#	Risk	Likelihood	Impact	Response
R1	Booking rush at rollout overloads the system	High	High	Scale only the Slot Booking Service, load test to 5,000 users (5.3)
R2	Two citizens get the last seat	Medium	High	Capacity check and decrement in one atomic step, regression test TC-07
R3	Certificate signing key leaks	Low	High	Key held only in the Certificate Service, rotate keys
R4	Health authority changes the interval rule	Medium	Medium	Interval is configuration, not code
R5	Personal data exposed in transit (BBPS-7)	Medium	High	TLS on every endpoint, security review before the pilot
R6	Certificate work under-estimated (seen in the Sprint 2 burndown)	Medium	Medium	Keep week 7 as buffer, split story 4.5 if it slips again